

# Explainable Financial Alerts for Retail Investors

## Integrating Statistical Anomaly Detection and News–Market Impact Correlation

Henrique J. S. Santos

Advisor: Prof. Luís Gomes   Co-advisor: Rafael Silva

MEIA, Instituto Superior de Engenharia do Porto

Defence

# The problem

- Most Americans own stock; many now trade through commission-free apps.
- The US equity market reached **US\$62.2 trillion** (end 2024).
- Retail investors face *continuous* price moves and news, **without** the tools of banks and funds.
- The AI spreading through finance is largely **opaque**, a poor fit for someone who must act and live with the result.

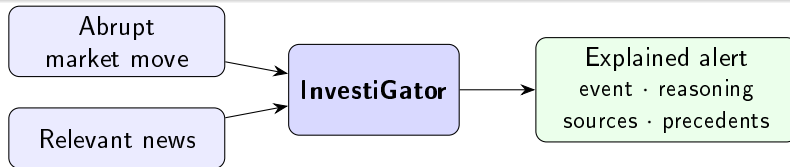
## Two distinct challenges

- 1 Tell a *genuinely* unusual move from ordinary volatility.
- 2 Read what a news item might *mean*, from how similar events behaved before.

# What InvestiGator does (and does not do)

It does *not* predict prices. It **explains**.

For each abrupt move or relevant news item, it shows *what* was detected, *how*, the *sources*, and which *past events* resemble it, with what happened to prices afterwards.



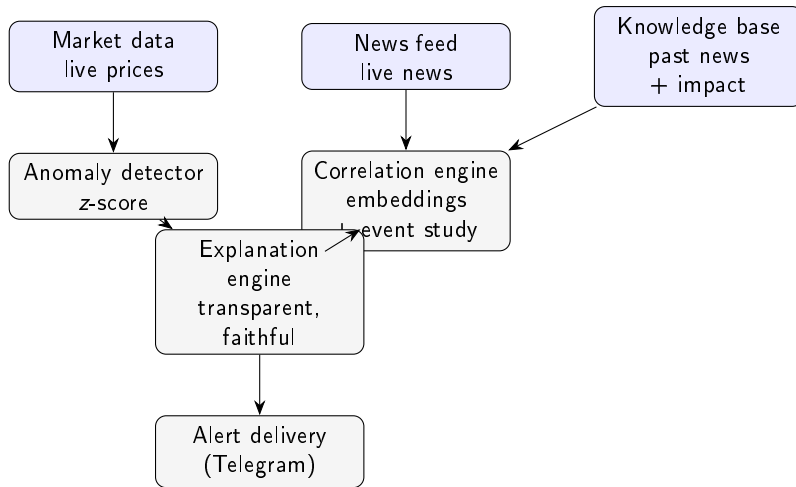
# Research questions

- RQ1** Can **transparent** statistical methods detect abrupt moves consistently enough to be useful, while staying explainable?
- RQ2** Can a news–market engine retrieve **genuinely analogous** precedents and quantify their impact **without lookahead**, better than trivial baselines?
- RQ3** Can the explanations be **faithful** to what the system computes, and useful to a retail investor?
- RQ4** Can a **supervised model**, trained on historical news and market context, usefully **prioritise** which news deserve an alert, beyond volatility baselines — **never** predicting direction?

## Contribution: Artificial Intelligence *Engineering*

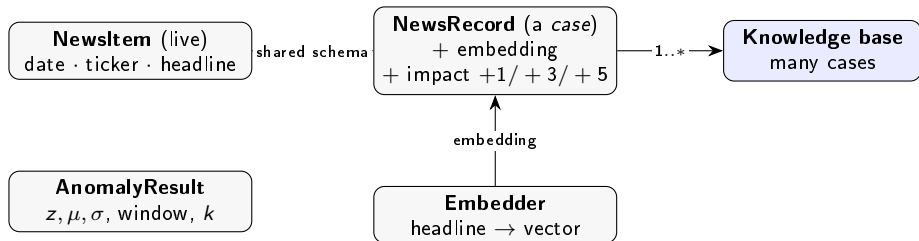
Not a new algorithm: the **disciplined integration, application and critical evaluation** of existing components — plus **one model trained by the author** (materiality triage) — into a working, explainable, reproducible system.

# Architecture



Single-responsibility components, common schema; two triggers converge on one explanation step.

# The data model — the objects



A *case* = one past headline + what happened to its price. Live **NewsItem** and historical **NewsRecord** share the schema, so a new headline is directly comparable.

# Market trigger: transparent by design

Rolling z-score, no lookahead:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \quad \text{flag if } |z_t| > k$$

$\mu_t, \sigma_t$  over the window *before*  $t$ . Every quantity is exposed to the explanation.

## Worked example (real)

TSLA, 24 Oct 2024 (post-earnings):

$\mu = -0.92\%$ ,  $\sigma = 2.73\%$ ,  $r = +19.8\%$

$\Rightarrow z = +7.61$  (flagged at  $k = 3$ ).

A fixed % threshold would treat a calm and a volatile stock alike; the z-score does not.

# News trigger: retrieve analogous precedents, measure impact

- 1 Embed the headline (Sentence-BERT).
- 2 Cosine similarity vs the knowledge base.
- 3 Return top- $k$  precedents.
- 4 Measure their impact at +1/ + 3/ + 5 days, from the event-day close (**no lookahead**).

*Case-based reasoning*: evidence, not a forecast.

## Worked example (real)

Query: "Nvidia demand surges on AI chip orders"

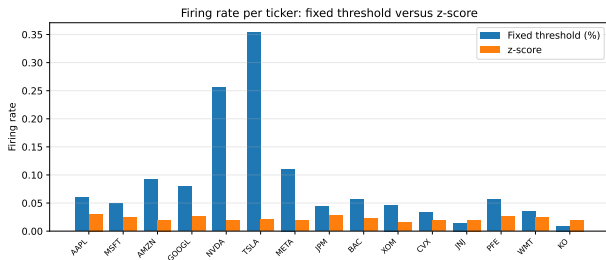
→ NVDA guidance (sim 0.60) +5d +3.5%

→ MSFT cloud-AI (sim 0.38) +5d +10.9%

→ NVDA accelerator (sim 0.38) +5d +4.9%

**Cross-ticker** match (MSFT); mean +5d  $\approx$  +6.5%.

# Result 1: anomaly detection is consistent across the market

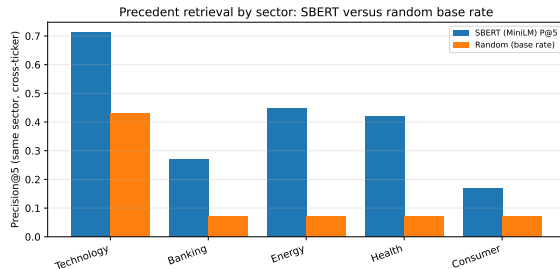


- Firing-rate range across 15 tickers: **0.015** (z-score) vs **0.344** (fixed %).
- $> 20\times$  more consistent, a **label-free** argument.
- Vs extreme-move proxy:  $F_1 = 0.516$  vs 0.218.
- Window ablation:  $F_1$  0.385 / 0.516 / 0.678 (10/20/60 d).

## Result 2: semantic retrieval beats every baseline

| Method         | P@5   | P@10  |
|----------------|-------|-------|
| SBERT (MiniLM) | 0.514 | 0.478 |
| SBERT (MPNet)  | 0.538 | 0.506 |
| Lexical        | 0.346 | 0.314 |
| Recency        | 0.126 | 0.106 |
| Random         | 0.240 | 0.240 |

Cross-ticker, 3,714 headlines, 5 seeds ( $\pm \sim 0.01$ ).



Lift largest where vocabulary is distinctive (energy, health).

## Result 3: explanations are faithful by construction

- Each alert is rendered *directly* from the computed objects.
- So the text **cannot diverge** from the computation.
- Checked by an automated test (alert reproduces each precedent's date/ticker/impact).
- Every alert ends with a no-prediction disclaimer.

### Real alert (news trigger)

```
News alert -- NVDA
"Nvidia raises guidance...AI data-centre"
Avg 1-day move:  -1.97% (over 5; 3 shown)
- MSFT (0.68) +1d -3.46%
- META (0.68) +1d -2.65%
- GOOGL (0.64) +1d -0.46%
observed past outcome, not a prediction
```

## Result 4: learned triage — trained, calibrated, honest

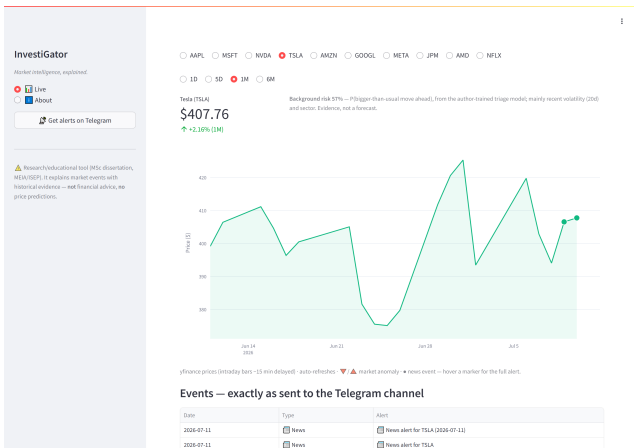
- **Task:**  $P(\text{abnormal move follows news})$  — materiality, **never** direction. Labels from our own event study ( $|\text{ticker} - \text{SPY}| \geq 2\%$  over  $(d, d+3]$ ).
- **Corpus:** FNSPID 2018–2023, 79,753 examples; temporal split by day + embargo; calibration on validation only.
- **Anti-lookahead:** a unit test *mutates future prices* — features unchanged, label changes.

| Model                | PR-AUC       | P@5/day      |
|----------------------|--------------|--------------|
| Always (floor)       | 0.378        | 0.163        |
| <b>Volatility LR</b> | <b>0.542</b> | <b>0.632</b> |
| Context LR           | 0.538        | 0.632        |
| Context+text LR      | 0.496        | 0.585        |
| Gradient boosting    | 0.469        | 0.551        |

### The honest verdict (pre-committed)

Within a 5-alerts/day budget, triage nearly **quadruples precision** (0.632 vs 0.163) — but **no text model beat volatility**. The signal is market context; reported as it fell.

# The product, live



- One page, a tab per watchlist ticker — a genuine screenshot, not a mockup.
- Background risk from the trained triage model, every day, no headline needed.
- Chart annotated with detected events + a history table — the **same** record the Telegram channel received.
- Free tier only: GitHub Actions (cron) + Streamlit Community Cloud.

## Limitations (stated plainly)

- Relevance is a **same-sector proxy**, not a human judgement.
- Retrieval corpus is a **recent** free-tier window; the *triage* study did use multi-year FNSPID (14/15 tickers — Meta is “FB” in the corpus).
- Displayed precedent impact is a **raw** post-event return (triage labels *are* market-adjusted).
- Similarity is **thematic, not directional**: a cluster can mix up/down moves, so the mean is theme-level *evidence*, not a forecast.
- **No human usefulness study** yet (RQ3 usefulness open).
- By design: **no price prediction, no trading**, so profit metrics are out of scope.

All results are reproducible from versioned scripts (re-run: identical numbers).

# Conclusions

- **RQ1 (yes):** transparent z-score detects consistently across the market and explains every flag.
- **RQ2 (yes, retrieval):** semantic retrieval finds analogous precedents far above baselines; impact measured without lookahead.
- **RQ3 (faithful yes; useful open):** explanations cannot diverge from the computation; human usefulness is future work.
- **RQ4 (no on text; yes on mechanism):** triage  $\approx 4\times$  within-budget precision, but volatility stayed unbeaten — the *second* fair “learned vs simple” test the transparent choice won (first: Isolation Forest vs z-score,  $F_1$  0.271 vs 0.530).

## Contribution

A working, **explainable, reproducible** financial-alert system for retail investors — existing components plus one author-trained triage model, engineered into a transparent whole, with an honest evaluation.

Thank you

# InvestiGator

Explainable financial alerts for retail investors

Questions welcome.

# Anticipated questions

- **Why not predict prices?** Market efficiency (Fama 1970): public news is impounded fast; we *measure* reactions, honestly.
- **Why z-score, not Isolation Forest/GARCH?** Transparency for a low-dimensional signal — and we *tested* it: a causal Isolation Forest on the same information lost ( $F_1$  0.271 vs 0.530).
- **Your trained model lost to the volatility baseline. Failure?** No: the comparison was *pre-committed* and answers RQ4 with evidence; triage still  $\approx 4\times$  within-budget precision vs alerting always, and the deployed variant uses exactly the features that carry the signal.
- **How do you know the triage model never sees the future?** A unit test mutates post-event prices and asserts features are unchanged while the label changes; temporal split by unique day with a five-day embargo.
- **Why precision@k and a sector proxy?** Standard ranked-retrieval metric; objective, reproducible; human relevance is future work.
- **Cross-ticker constraint?** An *evaluation* choice to avoid trivial self-matches; the live alert shows the most similar precedents regardless.
- **A positive headline retrieved negative precedents (−1.97%). Misleading?** Similarity matches *theme*, not direction; the mean is theme-level **evidence**, shown with every precedent and a no-prediction disclaimer. Future work: de-duplicate precedents and flag directional disagreement.
- **Is it reproducible?** Yes: fixed seeds, pinned deps, versioned scripts; re-running reproduces every number.